

APACHE  
**HBASE**





It can store huge amount of data in tabular format for extremely fast reads and writes

HBase is mostly used in a scenario that requires regular and consistent inserting and overwriting of data

HBase is a database management system designed in 2007 by Powerset, a Microsoft Company

# Why HBase when we have HDFS already?



HDFS stores, processes and manage large amounts of data efficiently. However, it performs only batch processing and the data will be accessed in a sequential manner.

Which means one have to insert an entire **dataset** for a simplest of job

Therefore, a **solution** is required to **access, read** or **write** data anytime regardless of its sequence in the clusters of data

## A table in hbase would look like



Each cell has multiple versions,  
typically represented by the timestamp  
of when they were inserted into the table

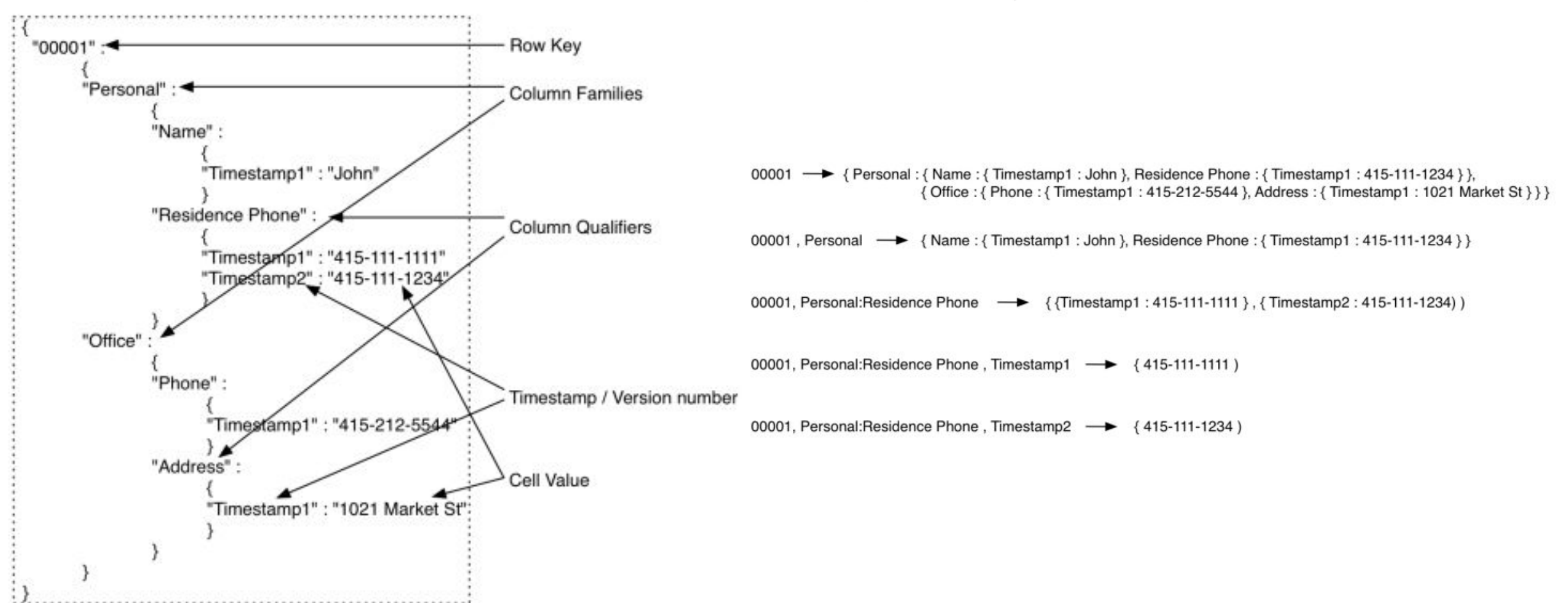
Timestamp1    Timestamp2

The table is lexicographically  
sorted on the row keys

| Row Key | Column Family - Personal |                 | Column Family - Office |                |
|---------|--------------------------|-----------------|------------------------|----------------|
|         | Name                     | Residence phone | Phone                  | Address        |
| 00001   | John                     | 415-111-1234    | 415-212-5544           | 1021 Market St |
| 00002   | Paul                     | 408-432-9922    | 415-212-5544           | 1021 Market St |
| 00003   | Ron                      | 415-993-2124    | 415-212-5544           | 1021 Market St |
| 00004   | Rob                      | 818-243-9988    | 408-998-4322           | 4455 Bird Ave  |
| 00005   | Carly                    | 206-221-9123    | 408-998-4325           | 4455 Bird Ave  |
| 00006   | Scott                    | 818-231-2566    | 650-443-2211           | 543 Dale Ave   |

Cells

# Lets not constrict our conceptualization to a tabular view



# Characteristic of HBase?

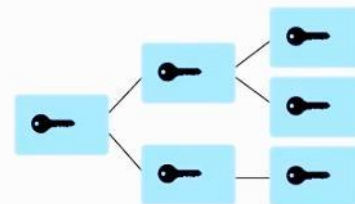
HBase is a type of NoSQL database and is classified as a key-value store



Value is identified  
with a key



Key and value are  
ByteArray



Values are stored  
in key-orders

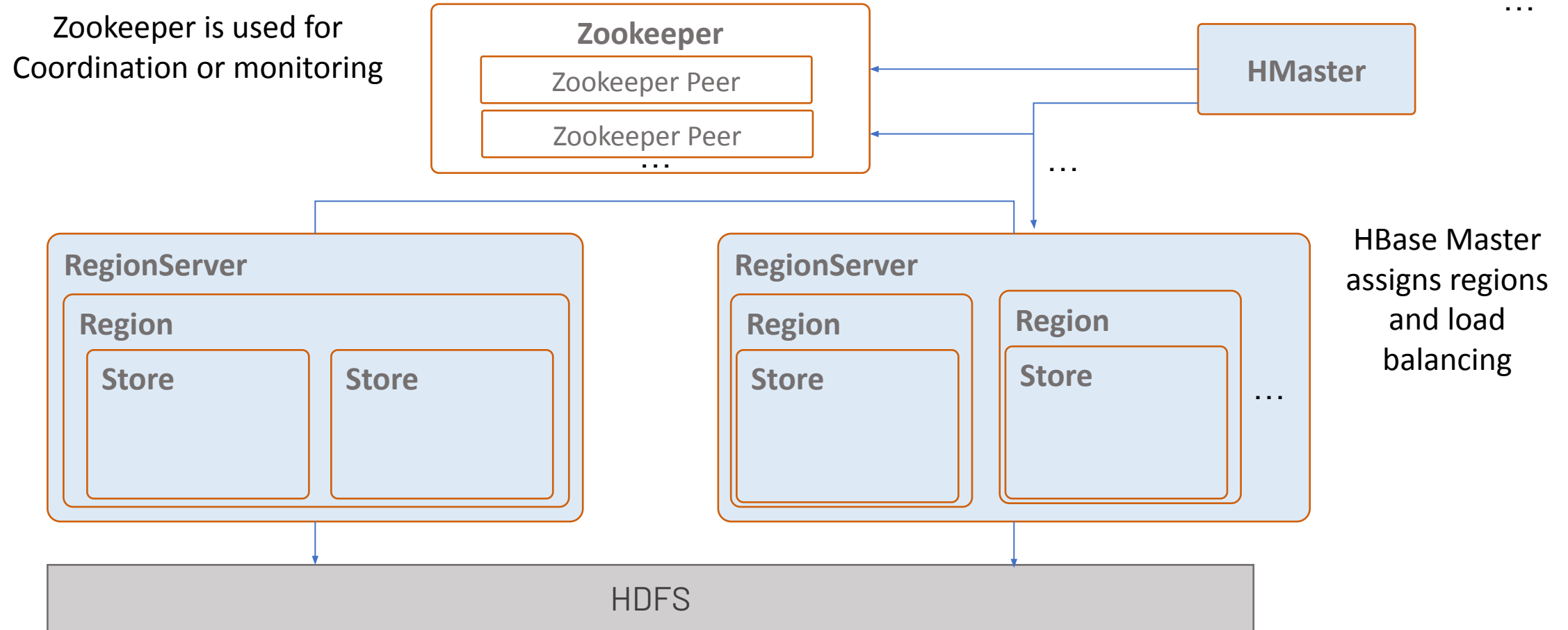


Quickly accessed  
by value keys

HBase is a database in which tables have no schema. At the time of table creation, columns families are defined, not columns

# HBase Components & Architecture

The HBase components include HBase Master and multiple RegionServers



## Storage Model of HBase

The two major components of the storage model are as follows:



### Partitioning:

- A table is horizontally partitioned into regions.
- Each region is managed by a RegionServer
- A RegionServer may hold multiple regions



### Persistence and data availability

- HBase stores its data in HDFS, does not replicate RegionServers, and relies on HDFS replication for data availability.
- Updates and reads are served from the in-memory cache called MemStore

The MemStore is a write buffer where HBase accumulates data in memory before a permanent write. Its contents are flushed to disk to form an HFile when the MemStore fills up. It doesn't write to an existing HFile but instead forms a new file on every flush. The HFile is the underlying storage format for HBase





Apache

**PIG**

## Why Pig ?

As we all know, Hadoop used MapReduce to analyze and process big data



Processing Big Data consumed more time

Processing Big Data was faster using MapReduce

## Why Pig ?

Prior to 2006, all **MapReduce** programs were written in Java

Non-programmers found it difficult to write lengthy Java codes



They faced issues in incorporating map, sort, reduce fundamentals of MapReduce while creating a program

Map Phase  
Shuffle and Sort  
Reduce phase



Eventually, it became a difficult task to maintain and optimize the code due to which the processing time increased



## Why Pig ?



Yahoo faced problems to process and analyze large datasets using Java as the codes were complex and lengthy

There was a necessity to develop an easier way to analyze large datasets without using time consuming complex Java codes

Problem

Necessity

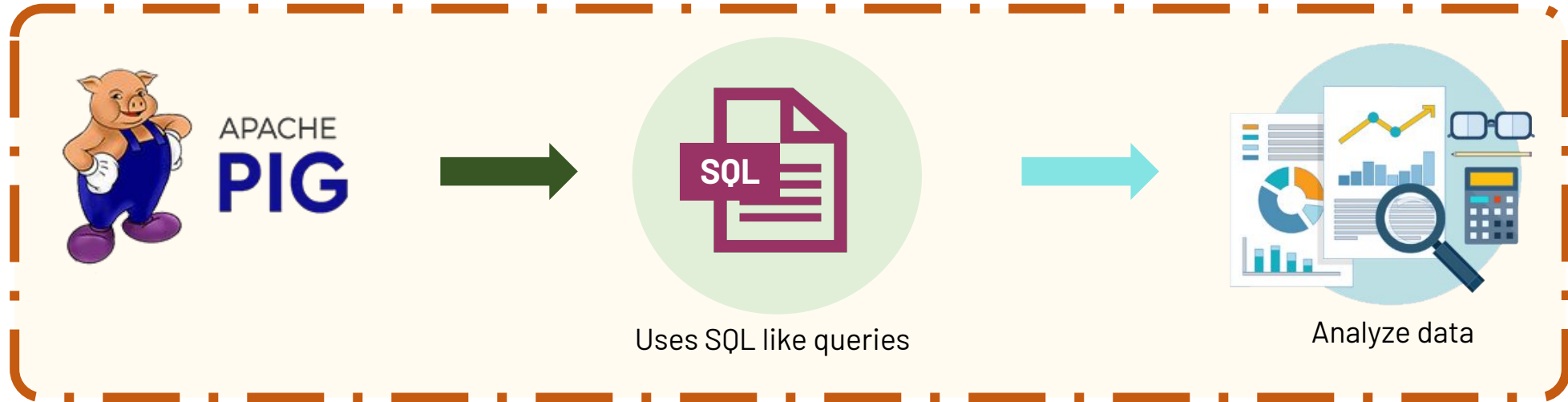


- Apache Pig was developed by Yahoo researchers.
- It was developed with a vision to analyze and process large datasets without using complex Java codes. Pig was developed especially for non-programmers
- Pig used simple steps to analyze datasets which was time efficient

Solution

## What is Pig ?

Pig is a scripting platform designed to process and analyze large data sets, as it runs on Hadoop cluster. Pig is extensible, self-optimizing, and easily programmed.



Pig operates on various types of data like structured, semi-structured and unstructured data

# Components of Pig



## Pig Latin

Pig Latin is the procedural data flow language used in Pig to analyze data

It is easy to program using Pig Latin as it is similar to SQL

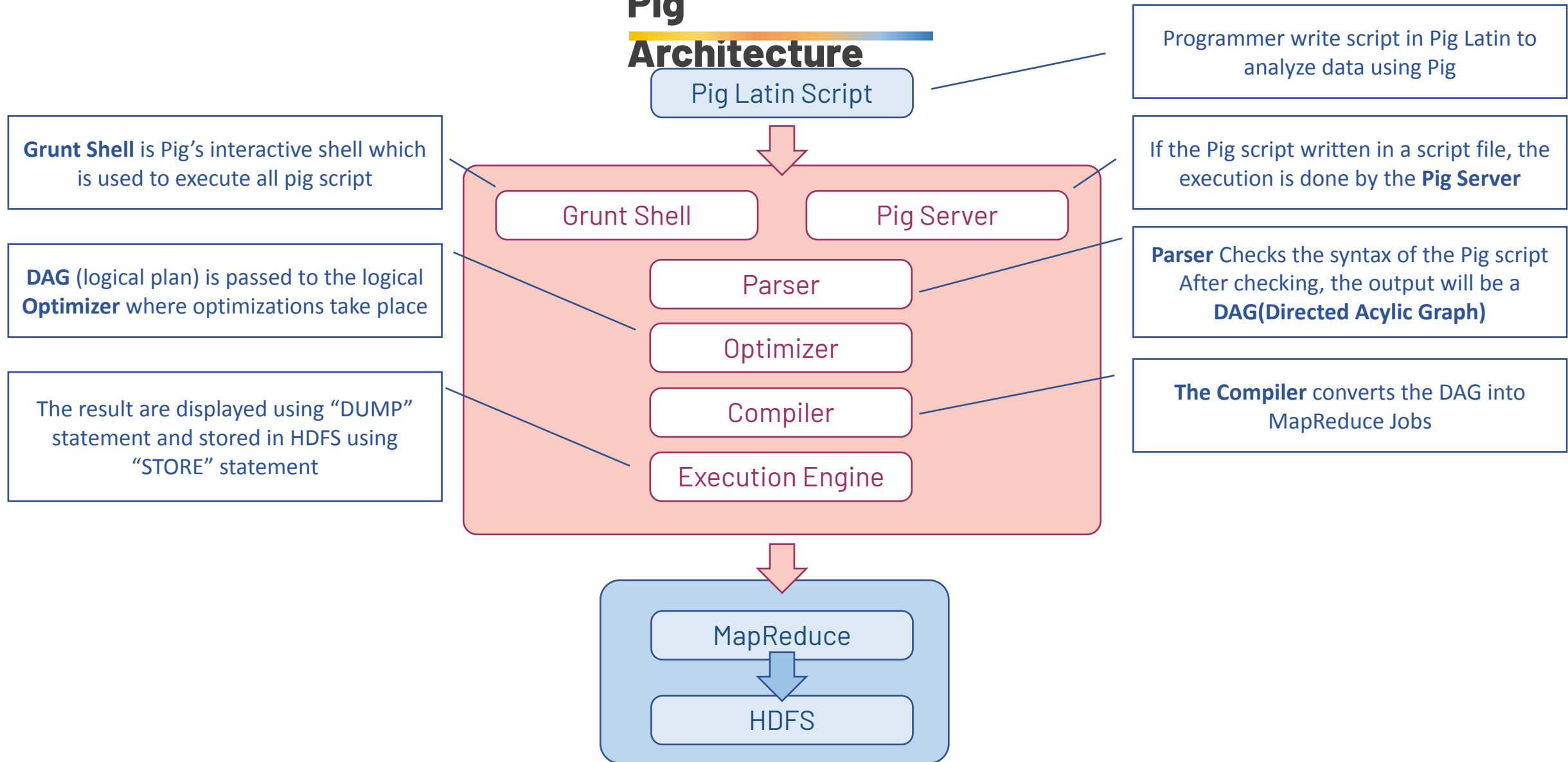
## Runtime Engine

Runtime engine represents the execution environment created to run Pig Latin programs

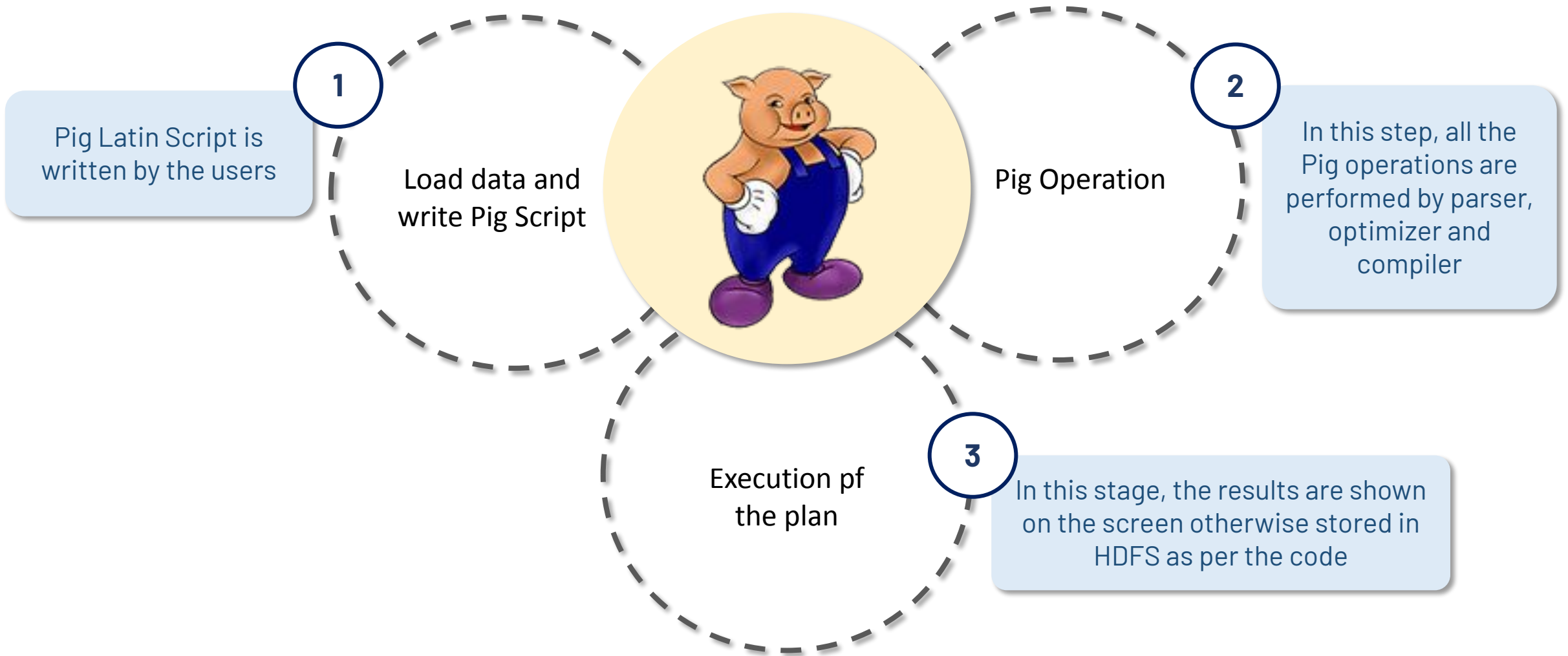
It is also a compiler that produces MapReduce programs

Used HDFS for storing and retrieving data

# Pig Architecture



# How Pig Works





# Pig Latin Data Model

As part of its data model, Pig supports four basic types

**Atom** represents any single value of primitive data type in Pig Latin like int, float, string. It is stored as string  
e.g. 'Rob' or 50

**Bag** is a collection of tuples. It is same as a table in RDBMS. It is represented by '{ }'  
e.g. { ( Rob , 5 ) , ( Mike , 10 ) }



**Tuple** represent sequence of fields that can be of any data type. It is same as a row in RDBMS i.e. a set of data from a single row  
e.g. ( Rob , 5 )

**Map** is a set of key-value pairs. Key is of chararray type and value can be of any type. It is represented by '[ ]'  
e.g. [ name # Mike , age # 10 ]

Pig Latin has a fully nestable data model that means one data type can be nested with another

# Pig Latin Data Model

Diagrammatical representation of the Pig Latin data model

| Sl.no | Name   | Age | Place       |
|-------|--------|-----|-------------|
| 01    | Gopal  | 35  | Fort Aguada |
| 02    | Laxman | 28  | Dona Paula  |
| 03    | Lucky  | 29  | GMC Complex |
| 04    | Madhav | 32  | Panaji      |

Diagrammatical representation of the Pig Latin data model showing a table with 4 columns (Sl.no, Name, Age, Place) and 4 rows. The first three rows are grouped by a bracket labeled 'Bag'. The last row is highlighted in yellow and is grouped by a bracket labeled 'Tuple'. The last row is also grouped by a bracket labeled 'Atom'.

# Pig Execution Mode

Pig in two execution modes. Depending on where the data is residing and where the pig script is going to run

Local mode

Here, the pig engine takes input from the Linux file system and the output is stored in the same file system

Local media is useful in analyzing small datasets using pig

MapReduce mode

Here, the Pig engine directly interacts and executes in HDFS and MapReduce.

In the MapReduce mode, queries written in Pig Latin are translated into MapReduce jobs and are run on a Hadoop Cluster. By Default, Pig runs on this mode

# Pig Execution Mode

There are three modes in Pig, depending on how a Pig Latin code can be written.



## Interactive Mode

Interactive mode means coding and executing the script, line by line



## Batch Mode

In Batch mode, all scripts are coded in a file with the extension .pig and the file is directly executed



## Embedded Mode

Pig lets its users define their own function (UDFs) in programming languages such as Java

## Use case - Twitter

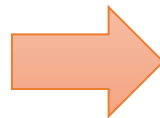
Analyze the user table and tweet table and find out how many tweets are created by a person

User Table

| ID | Name  |
|----|-------|
| 1  | Alice |
| 2  | Tim   |
| 3  | John  |

Tweet Table

| ID | Tweet         |
|----|---------------|
| 1  | Google....    |
| 2  | Tennis....    |
| 1  | Spacecraft... |
| 3  | Olympics...   |
| 1  | Politics...   |
| 2  | Oscar....     |



**First**, the twitter data is loaded onto the Pig storage using **LOAD** command

User Table

| ID | Name  |
|----|-------|
| 1  | Alice |
| 2  | Tim   |
| 3  | John  |

Tweet Table

| ID | Tweet         |
|----|---------------|
| 1  | Google....    |
| 2  | Tennis....    |
| 1  | Spacecraft... |
| 3  | Olympics...   |
| 1  | Politics...   |
| 2  | Oscar....     |

## Use case - Twitter

The remaining operation performed are shown below:

In **join** and **group** operation, the tweet and user tables are joined and grouped using **COGROUP** command

The next operation is the **aggregation**, the tweets are counted according to the names. The command used is **COUNT**

The result after the count operation is joined with the user table to find out the user name

| ID | Name  | Count |
|----|-------|-------|
| 1  | Alice | 3     |
| 2  | Tim   | 2     |
| 3  | John  | 1     |

| ID | Name  | Tweet         |
|----|-------|---------------|
| 1  | Alice | Google....    |
| 2  | Alice | Spacecraft... |
| 1  | Alice | Politics      |
| 3  | Tim   | Tennis....    |
| 1  | Tim   | Oscar....     |
| 2  | John  | Olympics...   |

Pig reduces the complexity of the operations which would have been lengthier using MapReduce

## Features of Pig

Ease of programming as Pig Latin is similar to SQL. Less lines of Code needs to be written

Short development time as the code is simpler

Handles all kind of data like structured, semi-structured and unstructured

Pig lets us create User-defined functions

Pig Offers a large set of operators such as join, filter as many more

Allows multiple queries to process parallelly

Optimization and compilation is easy as it is done automatically and internally







# Apache Hive

HIVE provides a SQL like interface for users to extract data from the Hadoop System

**Batch  
Processing**

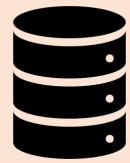


```
SELECT t1.a1 as c1, t2.b1 as c2 FROM t1  
join t2 ON ( t1.a2=t2.b2);
```

Resource Management

**Storage**

HDFS



HBase





# Features of Hive



Originally developed by Facebook around 2007



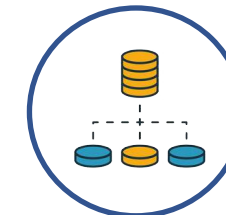
An open – source Apache project



High level abstraction layer on top of MapReduce and Apache Spark



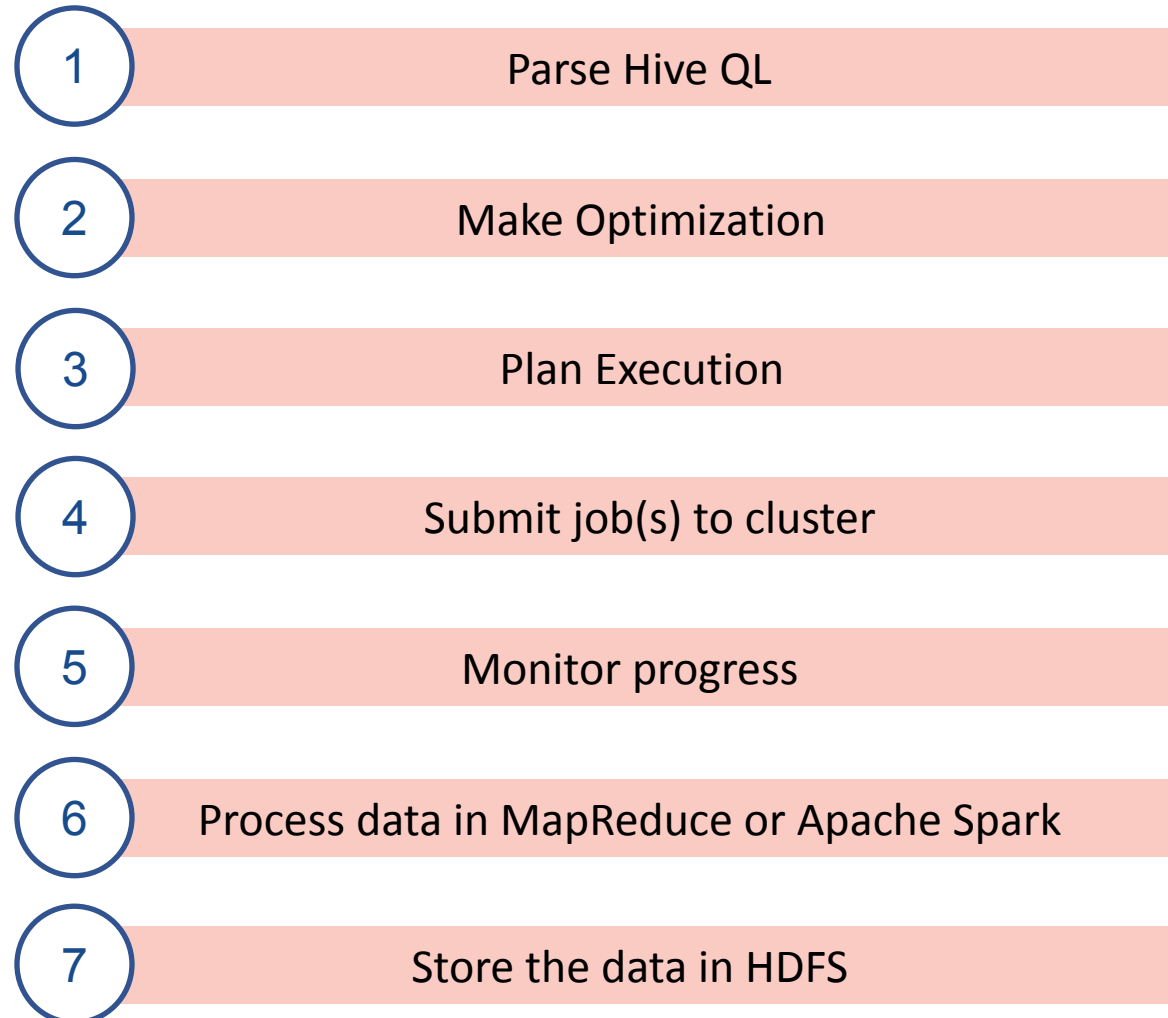
Uses HiveQL



Suitable for structure data



## Job Execution Flow in Hive



Watch Complete details here





# What is Apache Spark

?

Apache Spark is an open-source data processing engine to store and process data in real-time across various clusters of computers using simple programming constructs.

Supports Programming Language



# Hadoop vs Spark



Processing data using MapReduce is Slow

100 times faster than MapReduce as it is done in-memory

Performs batch processing of data

Batch processing and real-time processing of data

Since its written in Java. More lines of code and more execution time

Spark has fewer lines of code as it is implemented in **Scala**\*

*\*Both functional and OOP. Code are converted into bytecode and runs on JVM (Java Runtime Machine)*

# Spark

## Features

### Fast Processing

Spark contains **Resilient Distributed Datasets (RDD)** which saves time taken in reading and writing operation and hence, it runs almost ten to hundred times faster than Hadoop

### In-Memory Computing

In Spark, data is stored in the **RAM**, so it can access the data quickly and accelerate the speed of analytics

### Flexible

Spark supports **multiple languages** and allows the developers to write applications in Java, Scala, R or Python

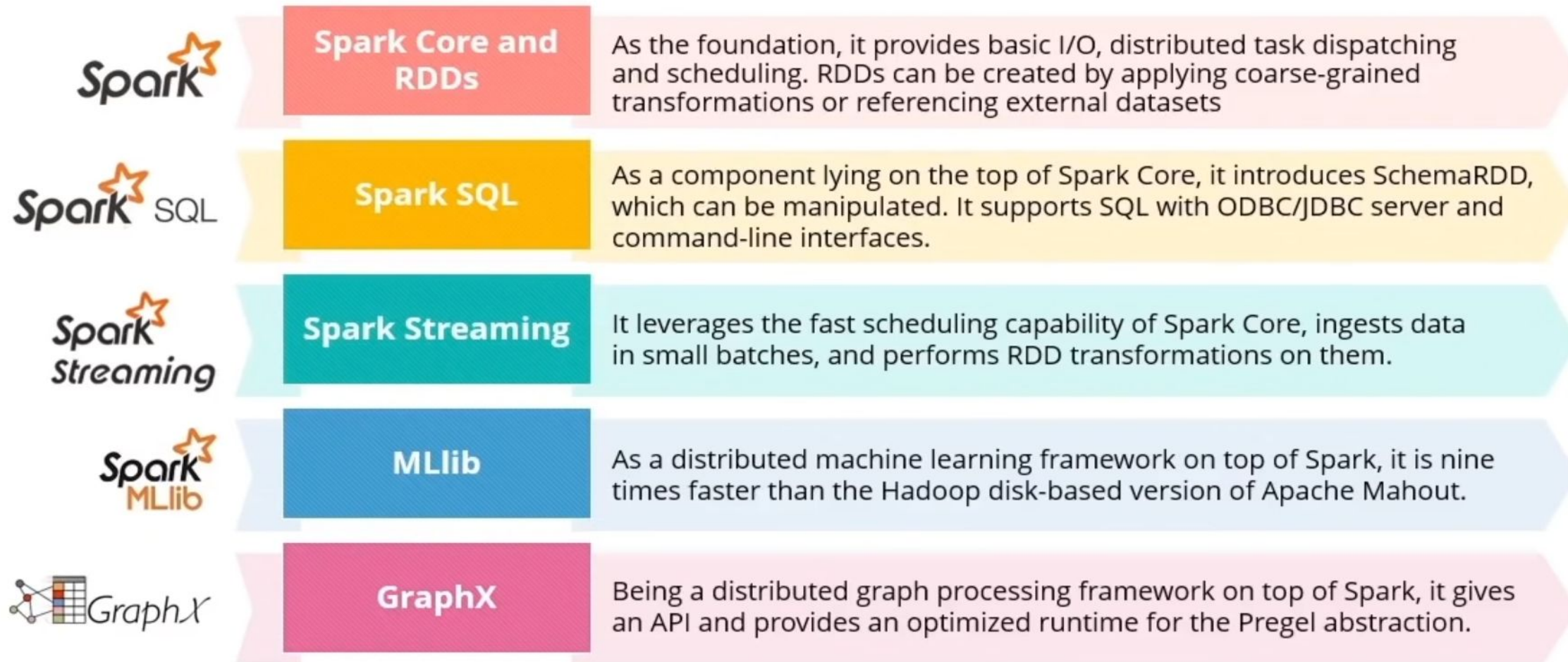
### Fault Tolerance

Spark contains **Resilient Distributed Datasets (RDD)** that are designed to handle the failure of any worker node in the cluster. Thus it ensures that the loss of data reduce to zero.

### Better Analytics

Spark has rich set of **SQL queries, machine learning algorithms, complex analytics**, etc. With all these functionalities, analytics can be performed better.

# Components of Spark





# Components of Spark



# Components of Spark

Spark Core is the base engine for large-scale parallel and distributed data processing

It is responsible for:



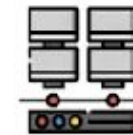
memory management



fault recovery



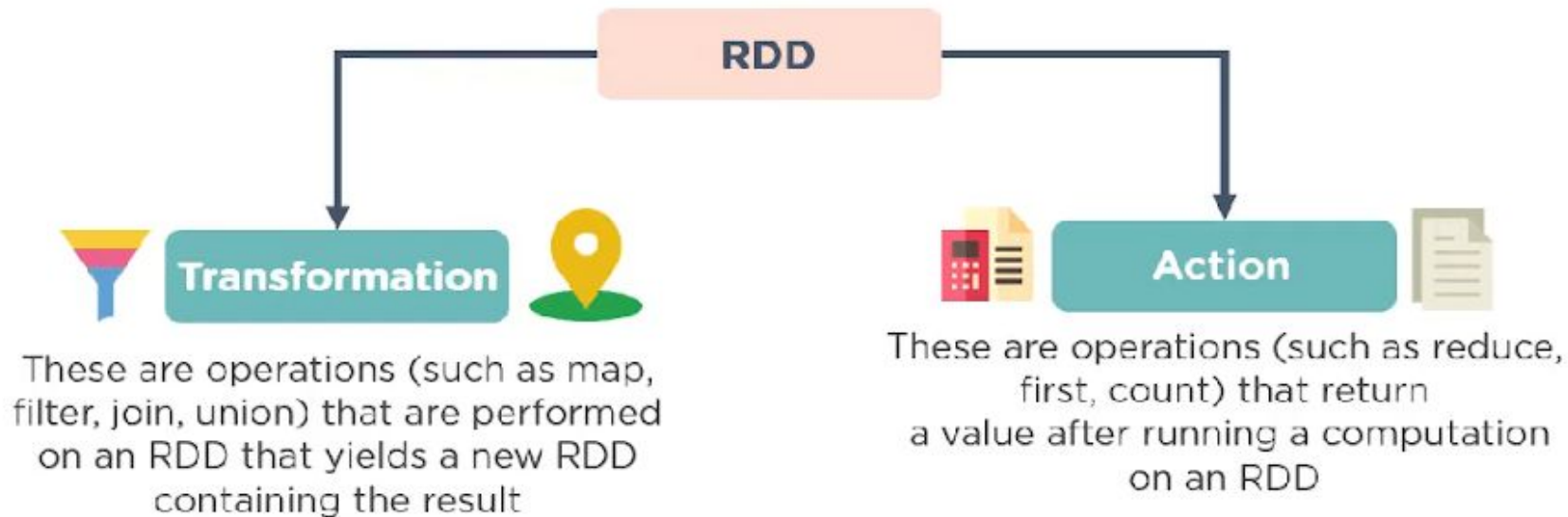
scheduling, distributing and  
monitoring jobs on a cluster



interacting with  
storage systems

# Components of Spark

Spark Core is embedded with RDDs (**Resilient Distributed Datasets**), an immutable fault-tolerant, distributed collection of objects that can be operated on in parallel



# Components of Spark

Spark SQL framework component is used for structured and semi-structured data processing

## Spark SQL Architecture

DataFrame DSL

Spark SQL and HQL

DataFrame API

Data Source API

CSV

JSON

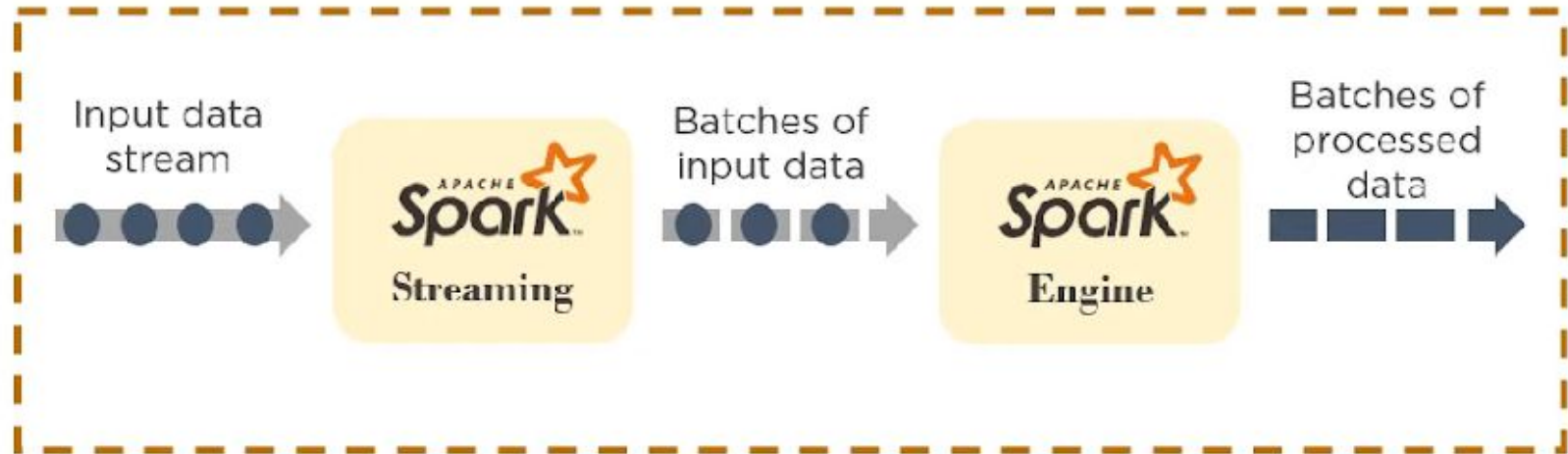
JDBC



# Components of Spark

Spark Streaming is a lightweight API that allows developers to perform batch processing and real-time streaming of data with ease

Provides secure, reliable, and fast processing of live data streams



# Components of Spark

MLlib is a low-level machine learning library that is simple to use, is scalable, and compatible with various programming languages

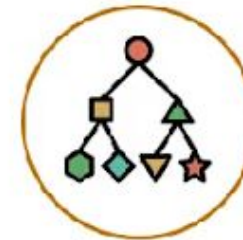
MLlib eases the deployment and development of scalable machine learning algorithms



It contains machine learning libraries that have an implementation of various machine learning algorithms



Clustering



Classification



Collaborative  
Filtering



# Components of Spark

GraphX is Spark's own Graph Computation Engine and data store



Provides a uniform tool for ETL



Exploratory data analysis

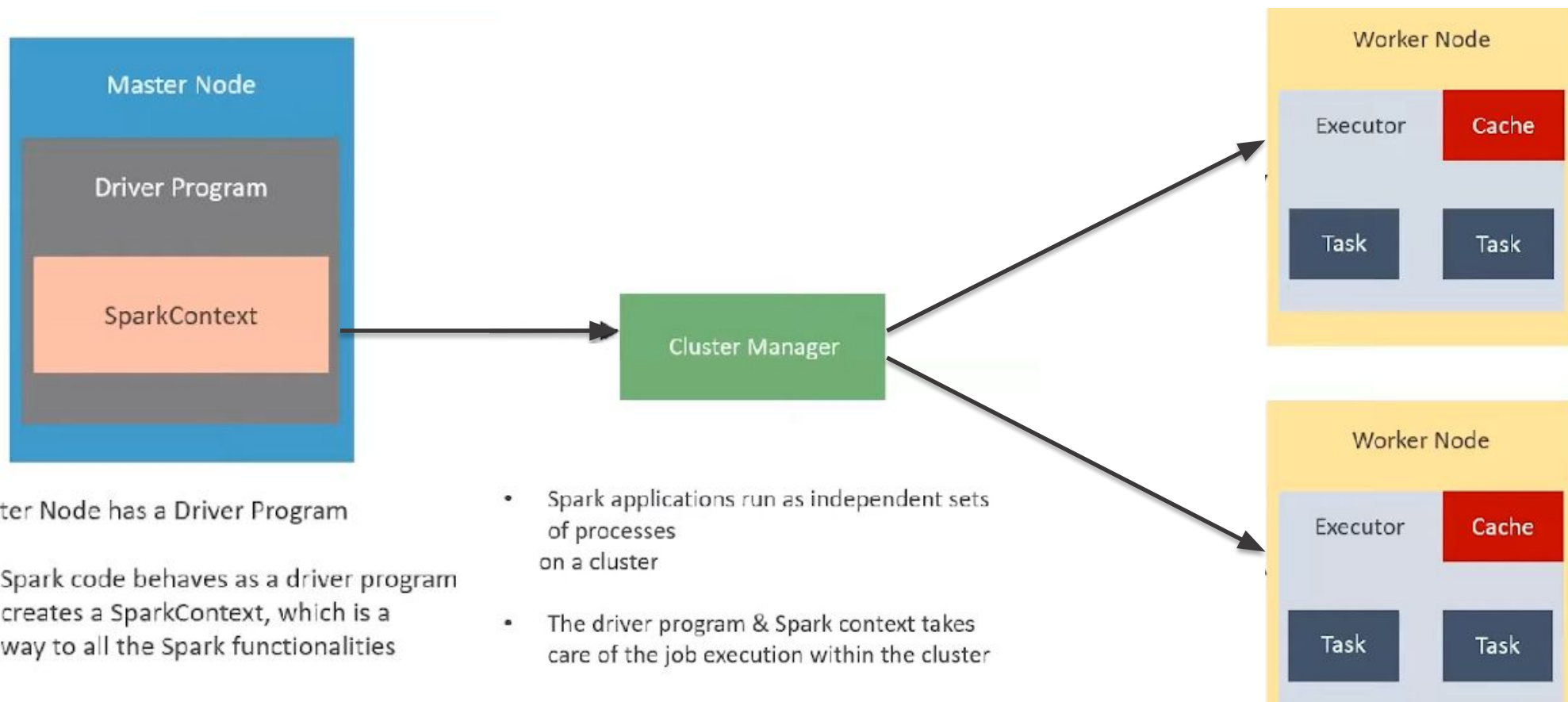


Interactive graph computations



# Components of

Apache Spark uses a master-slave architecture that consists of a driver, that runs on a master node, and multiple executors which run across the worker nodes in the cluster





## Spark - Advantage



The different advantages of Spark are:

- **Speed:** Extending the MapReduce model to support computation like stream processing and interactive queries
- **Combination:** Covering various workloads that used to require different distributed systems, which makes combining different processing types and allow easy tools management.
- **Hadoop support:** Allowing to create distributed datasets from any files stored in HDFS or any other support storage systems.

## Spark - Advantage

- Contains various closely-integrated components for distributing, scheduling and monitoring applications with many computational task
- **Empower** various higher-level components specialized for different workloads like machine learning for SQL.
- **Integrates** tightly allowing to create application that easily combine different processing models; For example, ability to write an application using Machine Learning to categories data in real times as it is ingestion from source of streaming
- Allows to access the same data through the Python shell for ad-hoc analysis and in standalone batch applications